Name:G.Eshwar H.No:2303A51808 Batch:26

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		**Assignment Type: Lab**	**Academic Year:**2025-2026
Course Coordinator Name	Dr. Rishabh Mittal		
Instructor(s) Name	Mr. S Naresh Kumar Ms. B. Swathi Dr. Sasanko Shekhar Gantayat Mr. Md Sallauddin Dr. Mathivanan Mr. Y Srikanth Ms. N Shilpa Dr. Rishabh Mittal (Coordinator) Dr. R. Prashant Kumar Mr. Ankushavali MD Mr. B Viswanath Ms. Sujitha Reddy Ms. A. Anitha Ms. M.Madhuri Ms. Katherashala Swetha Ms. Velpula sumalatha Mr. Bingi Raju		
CourseCode	23CS002PC304	**Course Title**	AI Assisted Coding
Year/Sem	III/II	**Regulation**	R23
Date and Day of Assignment	**Week8 – Wednesday**	**Time(s)**	23CSBTB01 To 23CSBTB52
Duration	2 Hours	**Applicable to Batches**	All batches
Assignment Number:16.3(Present assignment number)/**24**(Total number of assignments)			

Q.No.	Question	*Expected Time to complete*
1	**Lab 16 – Database Design and Queries: Schema Design and SQL Generation** **Lab Objectives:** • Understand schema design principles for relational databases.	Week8 - Wednesday

- Use AI-assisted tools to generate **SQL schema definitions**.
- Write and refine **SQL queries** (CRUD + aggregate operations).
- Explore how AI can optimize queries and ensure normalization.

Task 1 – Hospital Management Database

Task:

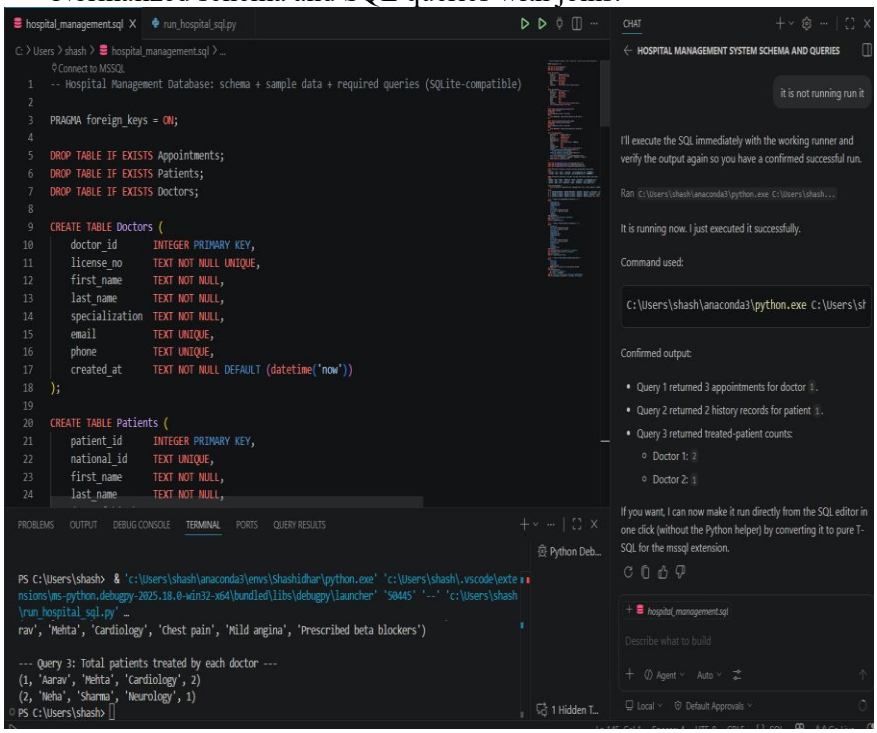
Create schema and queries for a **Hospital Management System**.

Instructions:

- Tables: Doctors, Patients, Appointments.
- Use AI to define constraints (unique IDs, valid dates).
- Generate queries:
  - List all appointments for a specific doctor.
  - Retrieve patient history by patient ID.
  - Count total patients treated by each doctor.

Expected Output:

- Normalized schema and SQL queries with joins.



hospital_management.sql | run_hospital_sql.py

C: > Users > shash > hospital_management.sql > ...

```
48
49  CREATE TABLE Appointments (
50      appointment_id      INTEGER PRIMARY KEY,
51      doctor_id           INTEGER NOT NULL,
52      patient_id          INTEGER NOT NULL,
53      appointment_start   TEXT NOT NULL,
54      appointment_end     TEXT NOT NULL,
55      status              TEXT NOT NULL DEFAULT 'scheduled',
56      reason              TEXT,
57      diagnosis           TEXT,
58      treatment_notes     TEXT,
59      created_at          TEXT NOT NULL DEFAULT (datetime('now')),
60      FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id)
61          ON UPDATE CASCADE ON DELETE RESTRICT,
62      FOREIGN KEY (patient_id) REFERENCES Patients(patient_id)
63          ON UPDATE CASCADE ON DELETE RESTRICT,
64      CHECK (datetime(appointment_end) > datetime(appointment_start)),
65      CHECK (status IN ('scheduled', 'completed', 'cancelled', 'no_show')),
66      UNIQUE (doctor_id, appointment_start)
67  );
68
69  CREATE INDEX idx_appointments_doctor_id ON Appointments(doctor_id);
70  CREATE INDEX idx_appointments_patient_id ON Appointments(patient_id);
71  CREATE INDEX idx_appointments_start ON Appointments(appointment_start);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

Python Deb...

```
PS C:\Users\shash>  & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\exte
nsions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '50445' '--' 'c:\Users\shash
\run_hospital_sql.py' …
rav', 'Mehta', 'Cardiology', 'Chest pain', 'Mild angina', 'Prescribed beta blockers')

--- Query 3: Total patients treated by each doctor ---
(1, 'Aarav', 'Mehta', 'Cardiology', 2)
(2, 'Neha', 'Sharma', 'Neurology', 1)
PS C:\Users\shash>
```

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hospital_management.sql | run_hospital_sql.py

C: > Users > shash > run_hospital_sql.py > ...

```
1   import sqlite3
2   from pathlib import Path
3
4   sql_path = Path(r"C:\Users\shash\hospital_management.sql")
5
6   conn = sqlite3.connect(":memory:")
7   cur = conn.cursor()
8
9   sql_text = sql_path.read_text(encoding="utf-8")
10
11  # Remove sqlite shell meta commands (e.g., .print) before executescript.
12  clean_lines = []
13  for line in sql_text.splitlines():
14      if line.strip().startswith("."):
15          continue
16      clean_lines.append(line)
17
18  cur.executescript("\n".join(clean_lines))
19
20  print("--- Query 1: All appointments for doctor_id = 1 ---")
21  for row in cur.execute(
22      """
23      SELECT
24          a.appointment_id,
25          a.appointment_start,
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

Python Deb...

```
PS C:\Users\shash>  & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\exte
nsions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '50445' '--' 'c:\Users\shash
\run_hospital_sql.py' …
rav', 'Mehta', 'Cardiology', 'Chest pain', 'Mild angina', 'Prescribed beta blockers')

--- Query 3: Total patients treated by each doctor ---
(1, 'Aarav', 'Mehta', 'Cardiology', 2)
(2, 'Neha', 'Sharma', 'Neurology', 1)
PS C:\Users\shash>
```

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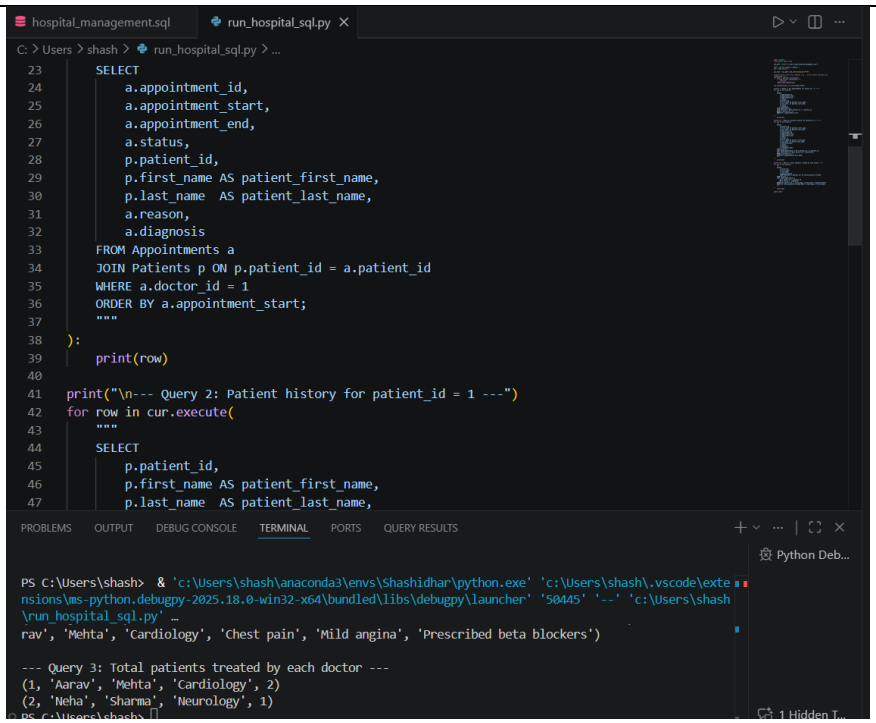
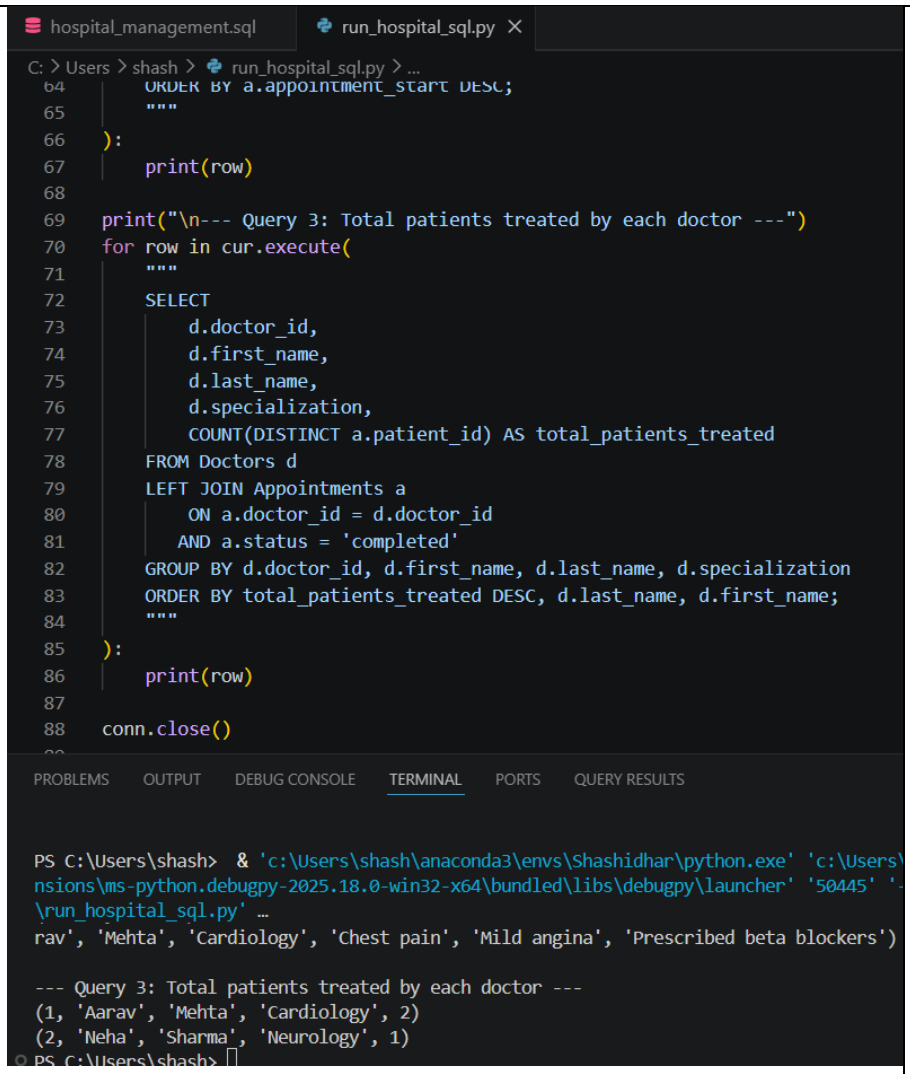

hospital_management.sql
run_hospital_sql.py
C: > Users > shash > run_hospital_sql.py > ...
SELECT
a.appointment_id,
a.appointment_start,
a.appointment_end,
a.status,
p.patient_id,
p.first_name AS patient_first_name,
p.last_name AS patient_last_name,
a.reason,
a.diagnosis
FROM Appointments a
JOIN Patients p ON p.patient_id = a.patient_id
WHERE a.doctor_id = 1
ORDER BY a.appointment_start;
"""
):
print(row)
print("\n--- Query 2: Patient history for patient_id = 1 ---")
for row in cur.execute(
"""
SELECT
p.patient_id,
p.first_name AS patient_first_name,
p.last_name AS patient_last_name,
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
QUERY RESULTS
Python Deb...
PS C:\Users\shash> & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '50445' '--' 'c:\Users\shash\run_hospital_sql.py' ...
rav', 'Mehta', 'Cardiology', 'Chest pain', 'Mild angina', 'Prescribed beta blockers')
--- Query 3: Total patients treated by each doctor ---
(1, 'Aarav', 'Mehta', 'Cardiology', 2)
(2, 'Neha', 'Sharma', 'Neurology', 1)
1 Hidden I...



Task 2 – Real-Time Application: Online Shopping Database

Scenario:

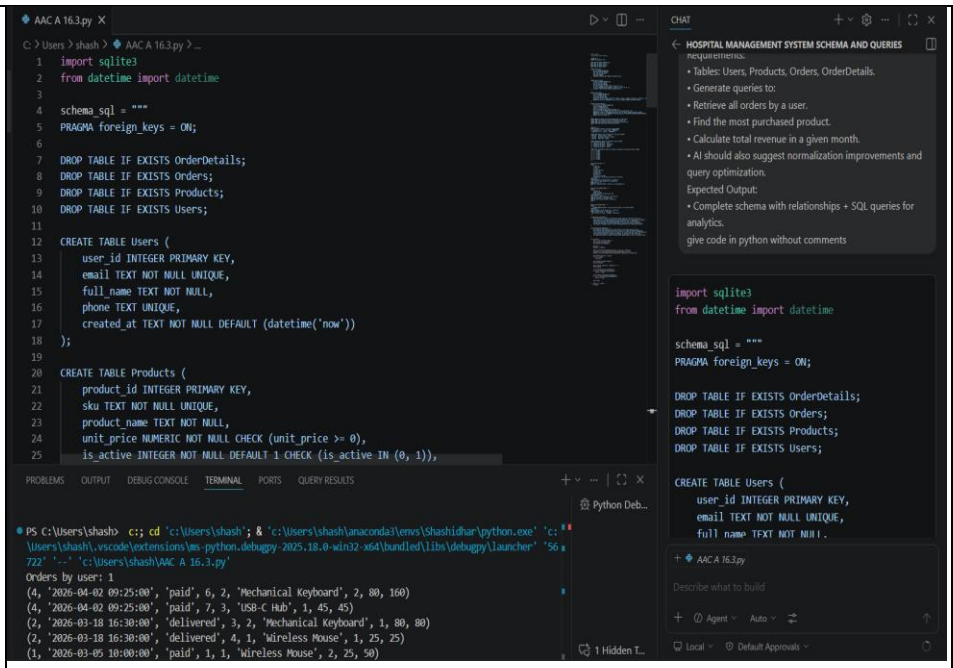
Design a database for an **E-commerce platform**.

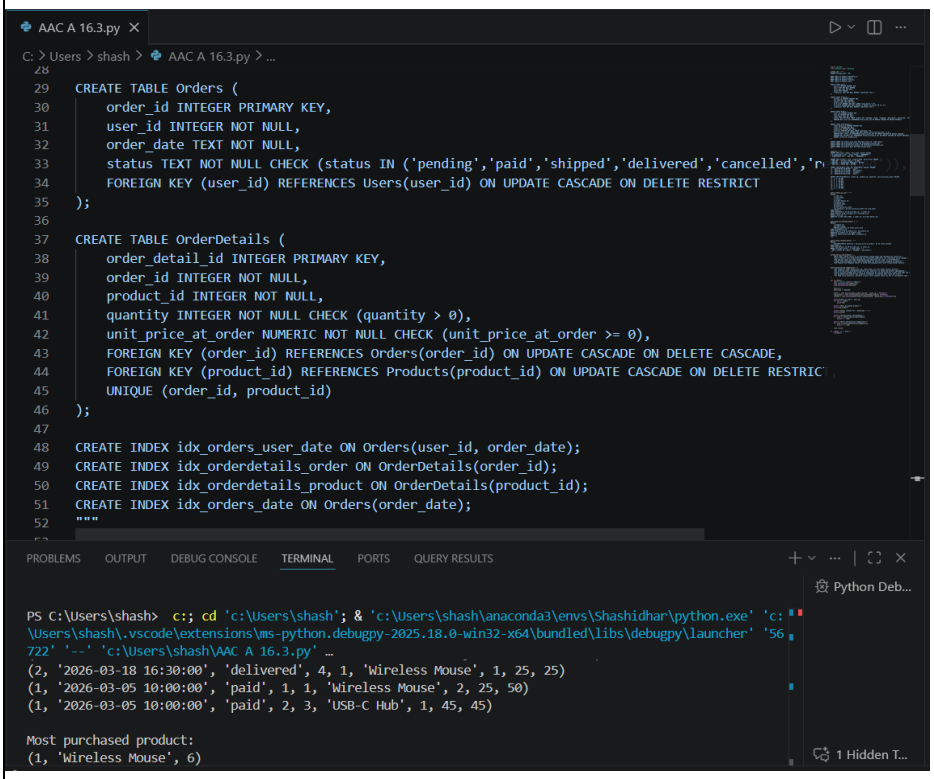
Requirements:

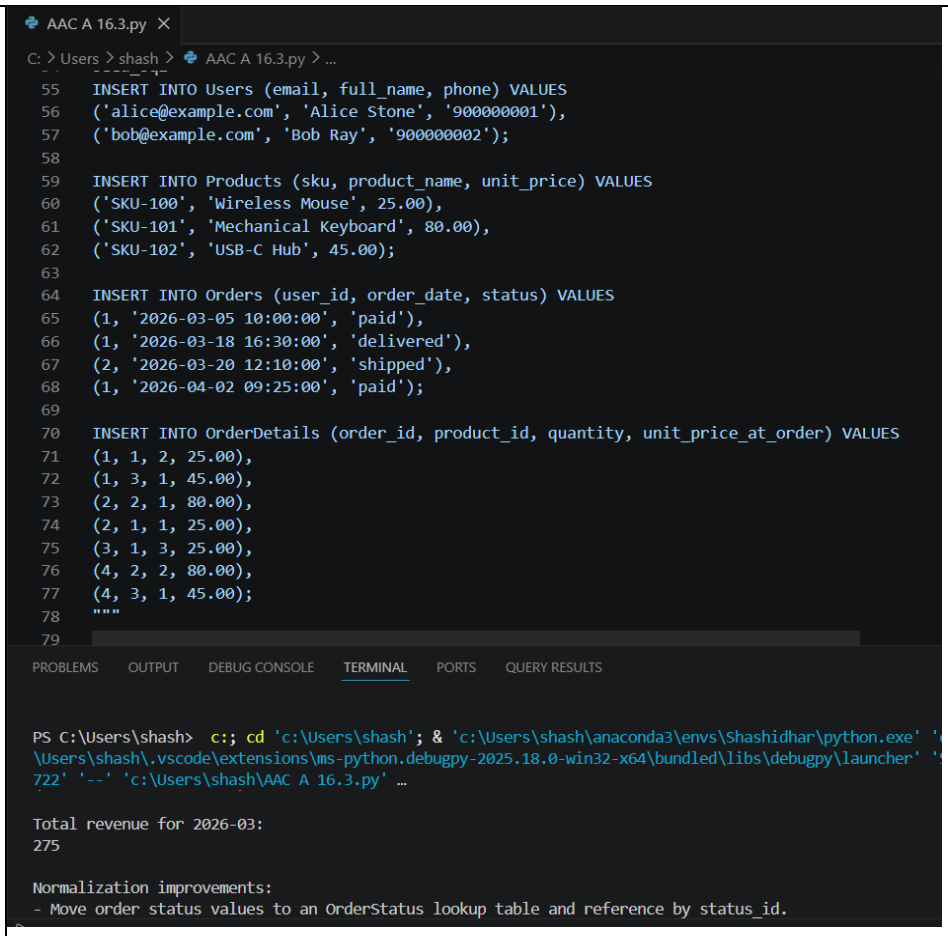
- Tables: Users, Products, Orders, OrderDetails.
- Generate queries to:
  - Retrieve all orders by a user.
  - Find the most purchased product.
  - Calculate total revenue in a given month.
- AI should also suggest **normalization improvements** and **query optimization**.

Expected Output:

- Complete schema with relationships + SQL queries for analytics.


AAC A 16.3.py
C: > Users > shash > AAC A 16.3.py > ...
import sqlite3
from datetime import datetime
schema_sql = """
PRAGMA foreign_keys = ON;
DROP TABLE IF EXISTS OrderDetails;
DROP TABLE IF EXISTS Orders;
DROP TABLE IF EXISTS Products;
DROP TABLE IF EXISTS Users;
CREATE TABLE Users (
user_id INTEGER PRIMARY KEY,
email TEXT NOT NULL UNIQUE,
full_name TEXT NOT NULL,
phone TEXT UNIQUE,
created_at TEXT NOT NULL DEFAULT (datetime('now'))
);
CREATE TABLE Products (
product_id INTEGER PRIMARY KEY,
sku TEXT NOT NULL UNIQUE,
product_name TEXT NOT NULL,
unit_price NUMERIC NOT NULL CHECK (unit_price >= 0),
is_active INTEGER NOT NULL DEFAULT 1 CHECK (is_active IN (0, 1)),
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS
Python Deb...
PS C:\Users\shash> c:; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:
\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '56
722' '--' 'c:\Users\shash\AAC A 16.3.py'
Orders by user: 1
(4, '2026-04-02 09:25:00', 'paid', 6, 2, 'Mechanical Keyboard', 2, 80, 160)
(4, '2026-04-02 09:25:00', 'paid', 7, 3, 'USB-C Hub', 1, 45, 45)
(2, '2026-03-18 16:30:00', 'delivered', 3, 2, 'Mechanical Keyboard', 1, 80, 80)
(2, '2026-03-18 16:30:00', 'delivered', 4, 1, 'Wireless Mouse', 1, 25, 25)
(1, '2026-03-05 10:00:00', 'paid', 1, 1, 'Wireless Mouse', 2, 25, 50)
1 Hidden T...
CHAT
HOSPITAL MANAGEMENT SYSTEM SCHEMA AND QUERIES
• Tables: Users, Products, Orders, OrderDetails.
• Generate queries to:
• Retrieve all orders by a user.
• Find the most purchased product.
• Calculate total revenue in a given month.
• AI should also suggest normalization improvements and query optimization.
Expected Output:
• Complete schema with relationships + SQL queries for analytics.
give code in python without comments
import sqlite3
from datetime import datetime
schema_sql = """
PRAGMA foreign_keys = ON;
DROP TABLE IF EXISTS OrderDetails;
DROP TABLE IF EXISTS Orders;
DROP TABLE IF EXISTS Products;
DROP TABLE IF EXISTS Users;
CREATE TABLE Users (
user_id INTEGER PRIMARY KEY,
email TEXT NOT NULL UNIQUE,
AAC A 16.3.py
Describe what to build
Agent
Auto
Local
Default Approvals


AAC A 16.3.py
C: > Users > shash > AAC A 16.3.py > ...
CREATE TABLE Orders (
order_id INTEGER PRIMARY KEY,
user_id INTEGER NOT NULL,
order_date TEXT NOT NULL,
status TEXT NOT NULL CHECK (status IN ('pending','paid','shipped','delivered','cancelled','r
FOREIGN KEY (user_id) REFERENCES Users(user_id) ON UPDATE CASCADE ON DELETE RESTRICT
);
CREATE TABLE OrderDetails (
order_detail_id INTEGER PRIMARY KEY,
order_id INTEGER NOT NULL,
product_id INTEGER NOT NULL,
quantity INTEGER NOT NULL CHECK (quantity > 0),
unit_price_at_order NUMERIC NOT NULL CHECK (unit_price_at_order >= 0),
FOREIGN KEY (order_id) REFERENCES Orders(order_id) ON UPDATE CASCADE ON DELETE CASCADE,
FOREIGN KEY (product_id) REFERENCES Products(product_id) ON UPDATE CASCADE ON DELETE RESTRIC
UNIQUE (order_id, product_id)
);
CREATE INDEX idx_orders_user_date ON Orders(user_id, order_date);
CREATE INDEX idx_orderdetails_order ON OrderDetails(order_id);
CREATE INDEX idx_orderdetails_product ON OrderDetails(product_id);
CREATE INDEX idx_orders_date ON Orders(order_date);
"""
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS
Python Deb...
PS C:\Users\shash> c:; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:
\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '56
722' '--' 'c:\Users\shash\AAC A 16.3.py' …
(2, '2026-03-18 16:30:00', 'delivered', 4, 1, 'Wireless Mouse', 1, 25, 25)
(1, '2026-03-05 10:00:00', 'paid', 1, 1, 'Wireless Mouse', 2, 25, 50)
(1, '2026-03-05 10:00:00', 'paid', 2, 3, 'USB-C Hub', 1, 45, 45)
Most purchased product:
(1, 'Wireless Mouse', 6)
1 Hidden T...


AAC A 16.3.py
C: > Users > shash > AAC A 16.3.py > ...
55 INSERT INTO Users (email, full_name, phone) VALUES
56 ('alice@example.com', 'Alice Stone', '900000001'),
57 ('bob@example.com', 'Bob Ray', '900000002');
58
59 INSERT INTO Products (sku, product_name, unit_price) VALUES
60 ('SKU-100', 'Wireless Mouse', 25.00),
61 ('SKU-101', 'Mechanical Keyboard', 80.00),
62 ('SKU-102', 'USB-C Hub', 45.00);
63
64 INSERT INTO Orders (user_id, order_date, status) VALUES
65 (1, '2026-03-05 10:00:00', 'paid'),
66 (1, '2026-03-18 16:30:00', 'delivered'),
67 (2, '2026-03-20 12:10:00', 'shipped'),
68 (1, '2026-04-02 09:25:00', 'paid');
69
70 INSERT INTO OrderDetails (order_id, product_id, quantity, unit_price_at_order) VALUES
71 (1, 1, 2, 25.00),
72 (1, 3, 1, 45.00),
73 (2, 2, 1, 80.00),
74 (2, 1, 1, 25.00),
75 (3, 1, 3, 25.00),
76 (4, 2, 2, 80.00),
77 (4, 3, 1, 45.00);
78 """
79
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS
PS C:\Users\shash> c:; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe'
\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher'
722' '--' 'c:\Users\shash\AAC A 16.3.py' …
Total revenue for 2026-03:
275
Normalization improvements:
- Move order status values to an OrderStatus lookup table and reference by status_id.


AAC A 16.3.py
C: > Users > shash > AAC A 16.3.py > ...
query_orders_by_user = """
SELECT
o.order_id,
o.order_date,
o.status,
od.order_detail_id,
p.product_id,
p.product_name,
od.quantity,
od.unit_price_at_order,
(od.quantity * od.unit_price_at_order) AS line_total
FROM Orders o
JOIN OrderDetails od ON od.order_id = o.order_id
JOIN Products p ON p.product_id = od.product_id
WHERE o.user_id = ?
ORDER BY o.order_date DESC, o.order_id, od.order_detail_id;
"""
query_most_purchased_product = """
SELECT
p.product_id,
p.product_name,
SUM(od.quantity) AS total_units_sold
FROM OrderDetails od
JOIN Products p ON p.product_id = od.product_id
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
QUERY RESULTS
PS C:\Users\shash> c:; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c
\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '5
722' '--' 'c:\Users\shash\AAC A 16.3.py' ...
Normalization improvements:
- Move order status values to an OrderStatus lookup table and reference by status_id.
- Split Products into ProductCatalog and ProductPricingHistory for time-based price changes.
- Add Address table and keep shipping/billing addresses per order in OrderAddress.
- Create ProductCategory and ProductCategoryMap for many-to-many category assignments.
- Add Payment and Shipment tables to avoid overloading Orders with transactional details.

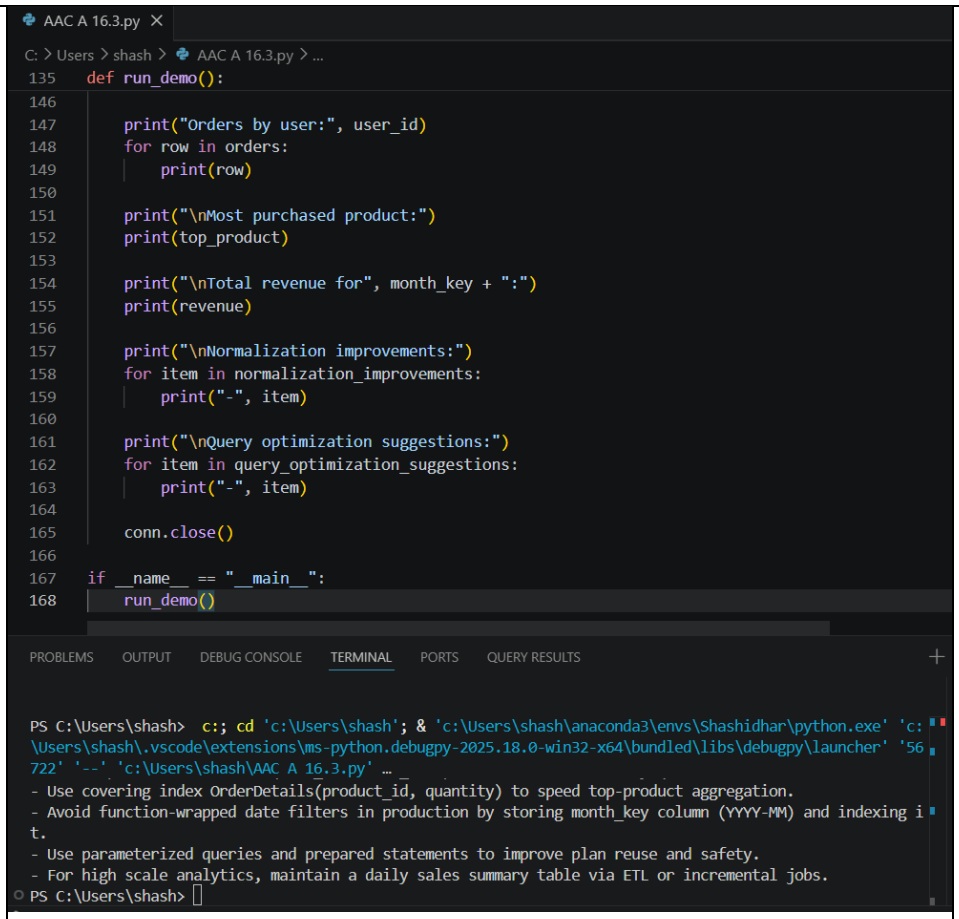
AAC A 16.3.py

C: > Users > shash > AAC A 16.3.py > ...

```
110    query_total_revenue_month = """
111    SELECT
112        COALESCE(SUM(od.quantity * od.unit_price_at_order), 0) AS total_revenue
113    FROM Orders o
114    JOIN OrderDetails od ON od.order_id = o.order_id
115    WHERE strftime('%Y-%m', o.order_date) = ?
116      AND o.status IN ('paid', 'shipped', 'delivered');
117    """
118
119    normalization_improvements = [
120        "Move order status values to an OrderStatus lookup table and reference by status_id.",
121        "Split Products into ProductCatalog and ProductPricingHistory for time-based price changes
122        "Add Address table and keep shipping/billing addresses per order in OrderAddress.",
123        "Create ProductCategory and ProductCategoryMap for many-to-many category assignments.",
124        "Add Payment and Shipment tables to avoid overloading Orders with transactional details."
125    ]
126
127    query_optimization_suggestions = [
128        "Use composite index Orders(user_id, order_date) for user-order history queries.",
129        "Use covering index OrderDetails(product_id, quantity) to speed top-product aggregation.",
130        "Avoid function-wrapped date filters in production by storing month_key column (YYYY-MM) a
131        "Use parameterized queries and prepared statements to improve plan reuse and safety.",
132        "For high scale analytics, maintain a daily sales summary table via ETL or incremental job
133    ]
134
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

```
PS C:\Users\shash>  c:; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:
\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '56
722' '--' 'c:\Users\shash\AAC A 16.3.py' …
Query optimization suggestions:
- Use composite index Orders(user_id, order_date) for user-order history queries.
- Use covering index OrderDetails(product_id, quantity) to speed top-product aggregation.
- Avoid function-wrapped date filters in production by storing month_key column (YYYY-MM) and indexing i
t.
- Use parameterized queries and prepared statements to improve plan reuse and safety.
```



Task 3 – Library Database

Task:

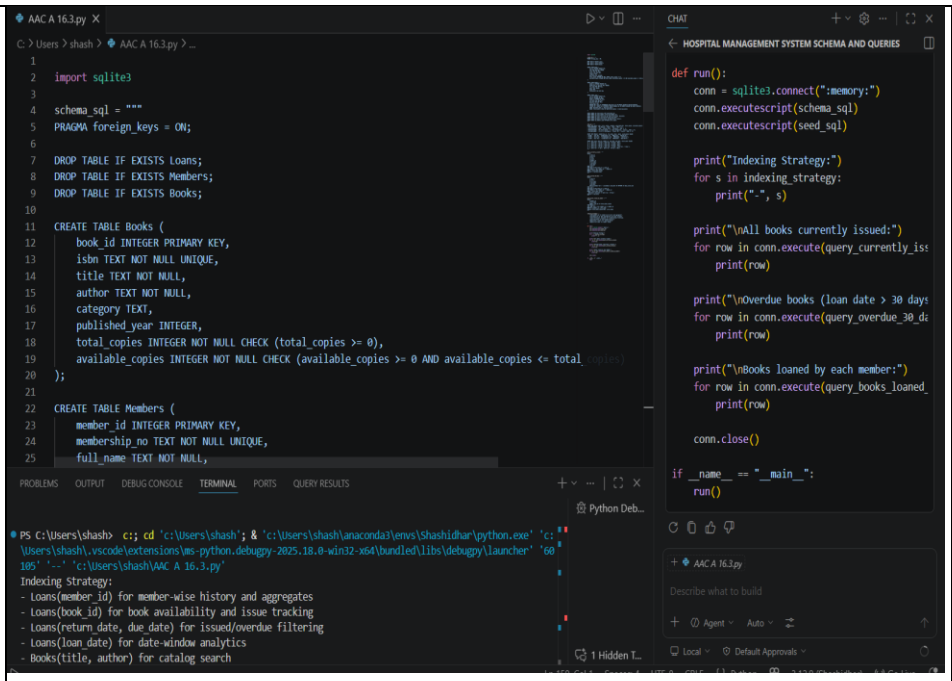
Design schema for a **Library Management System**.

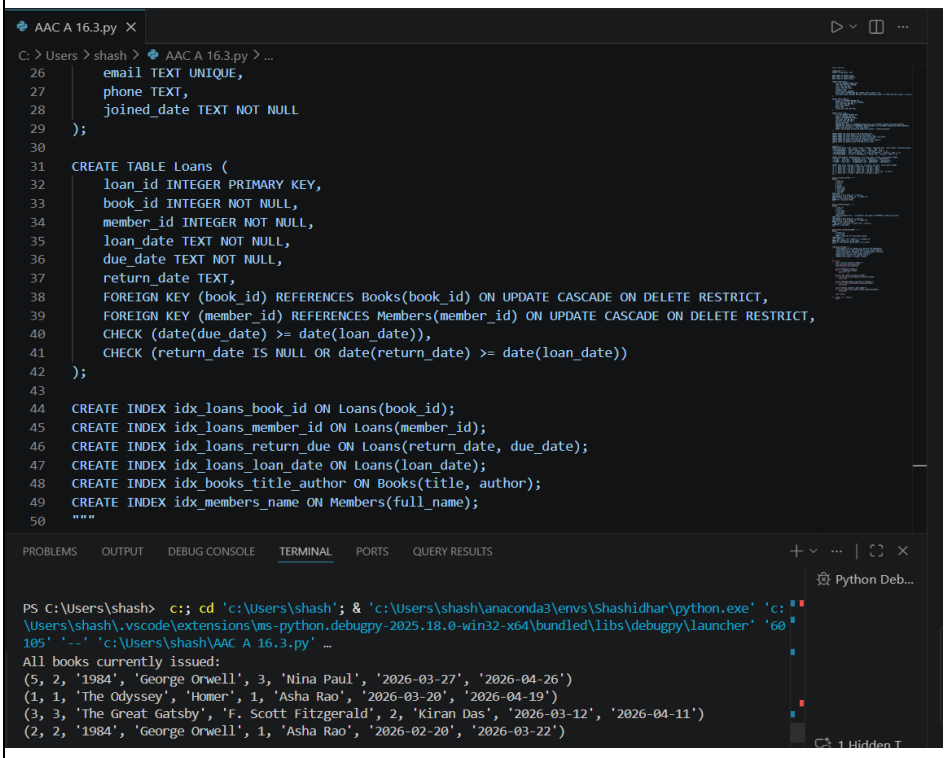
Instructions:

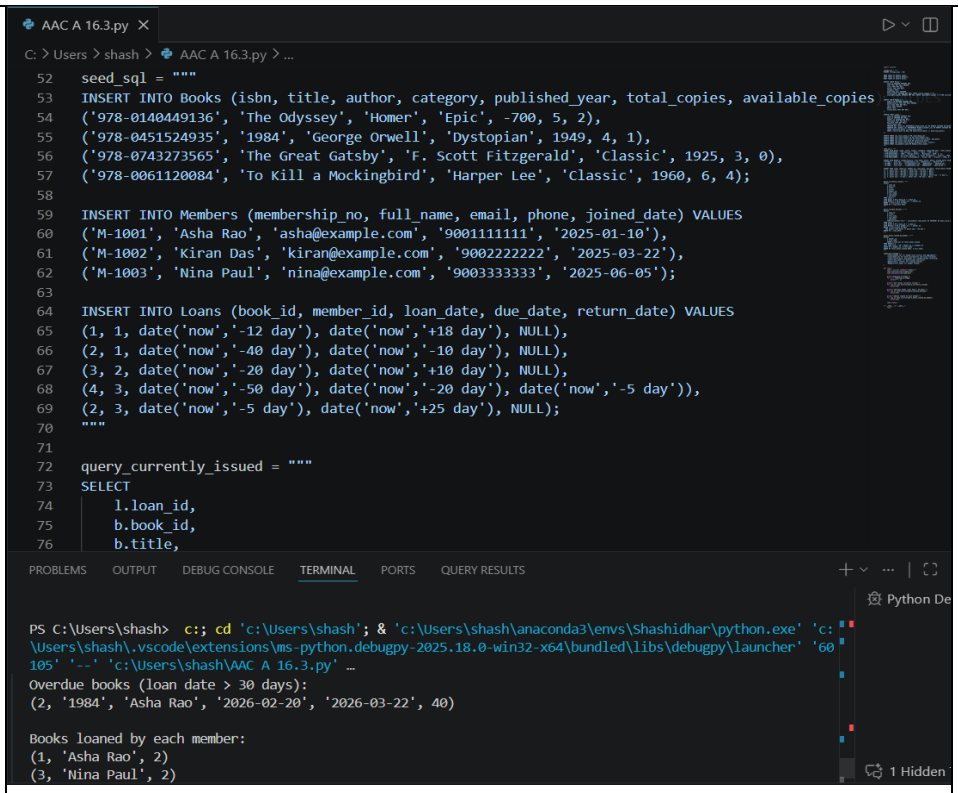
- Tables: Books, Members, Loans.
- Use AI to suggest indexing strategy for faster queries.
- Generate queries:
  - Retrieve all books currently issued.
  - Find overdue books (loan date > 30 days).
  - Count number of books loaned by each member.

Expected Output:

- Schema + SQL queries demonstrating joins and conditions.


AAC A 16.3.py
C: > Users > shash > AAC A 16.3.py > ...
import sqlite3
schema_sql = """
PRAGMA foreign_keys = ON;
DROP TABLE IF EXISTS Loans;
DROP TABLE IF EXISTS Members;
DROP TABLE IF EXISTS Books;
CREATE TABLE Books (
book_id INTEGER PRIMARY KEY,
isbn TEXT NOT NULL UNIQUE,
title TEXT NOT NULL,
author TEXT NOT NULL,
category TEXT,
published_year INTEGER,
total_copies INTEGER NOT NULL CHECK (total_copies >= 0),
available_copies INTEGER NOT NULL CHECK (available_copies >= 0 AND available_copies <= total
);
CREATE TABLE Members (
member_id INTEGER PRIMARY KEY,
membership_no TEXT NOT NULL UNIQUE,
full_name TEXT NOT NULL,
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS
Python Deb...
PS C:\Users\shash> c:; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:
\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '60
105' '--' 'c:\Users\shash\AAC A 16.3.py'
Indexing Strategy:
- Loans(member_id) for member-wise history and aggregates
- Loans(book_id) for book availability and issue tracking
- Loans(return_date, due_date) for issued/overdue filtering
- Loans(loan_date) for date-window analytics
- Books(title, author) for catalog search
1 Hidden T...
CHAT
HOSPITAL MANAGEMENT SYSTEM SCHEMA AND QUERIES
def run():
conn = sqlite3.connect(":memory:")
conn.executescript(schema_sql)
conn.executescript(seed_sql)
print("Indexing Strategy:")
for s in indexing_strategy:
print("-", s)
print("\nAll books currently issued:")
for row in conn.execute(query_currently_iss
print(row)
print("\nOverdue books (loan date > 30 days
for row in conn.execute(query_overdue_30_da
print(row)
print("\nBooks loaned by each member:")
for row in conn.execute(query_books_loaned_
print(row)
conn.close()
if __name__ == "__main__":
run()
AAC A 16.3.py
Describe what to build
Agent
Auto
Local
Default Approvals


AAC A 16.3.py
C: > Users > shash > AAC A 16.3.py > ...
email TEXT UNIQUE,
phone TEXT,
joined_date TEXT NOT NULL
);
CREATE TABLE Loans (
loan_id INTEGER PRIMARY KEY,
book_id INTEGER NOT NULL,
member_id INTEGER NOT NULL,
loan_date TEXT NOT NULL,
due_date TEXT NOT NULL,
return_date TEXT,
FOREIGN KEY (book_id) REFERENCES Books(book_id) ON UPDATE CASCADE ON DELETE RESTRICT,
FOREIGN KEY (member_id) REFERENCES Members(member_id) ON UPDATE CASCADE ON DELETE RESTRICT,
CHECK (date(due_date) >= date(loan_date)),
CHECK (return_date IS NULL OR date(return_date) >= date(loan_date))
);
CREATE INDEX idx_loans_book_id ON Loans(book_id);
CREATE INDEX idx_loans_member_id ON Loans(member_id);
CREATE INDEX idx_loans_return_due ON Loans(return_date, due_date);
CREATE INDEX idx_loans_loan_date ON Loans(loan_date);
CREATE INDEX idx_books_title_author ON Books(title, author);
CREATE INDEX idx_members_name ON Members(full_name);
"""
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS
Python Deb...
PS C:\Users\shash> c:; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:
\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '60
105' '--' 'c:\Users\shash\AAC A 16.3.py' ...
All books currently issued:
(5, 2, '1984', 'George Orwell', 3, 'Nina Paul', '2026-03-27', '2026-04-26')
(1, 1, 'The Odyssey', 'Homer', 1, 'Asha Rao', '2026-03-20', '2026-04-19')
(3, 3, 'The Great Gatsby', 'F. Scott Fitzgerald', 2, 'Kiran Das', '2026-03-12', '2026-04-11')
(2, 2, '1984', 'George Orwell', 1, 'Asha Rao', '2026-02-20', '2026-03-22')
1 Hidden T...


AAC A 16.3.py
C: > Users > shash > AAC A 16.3.py > ...
seed_sql = """
INSERT INTO Books (isbn, title, author, category, published_year, total_copies, available_copies
('978-0140449136', 'The Odyssey', 'Homer', 'Epic', -700, 5, 2),
('978-0451524935', '1984', 'George Orwell', 'Dystopian', 1949, 4, 1),
('978-0743273565', 'The Great Gatsby', 'F. Scott Fitzgerald', 'Classic', 1925, 3, 0),
('978-0061120084', 'To Kill a Mockingbird', 'Harper Lee', 'Classic', 1960, 6, 4);
INSERT INTO Members (membership_no, full_name, email, phone, joined_date) VALUES
('M-1001', 'Asha Rao', 'asha@example.com', '9001111111', '2025-01-10'),
('M-1002', 'Kiran Das', 'kiran@example.com', '9002222222', '2025-03-22'),
('M-1003', 'Nina Paul', 'nina@example.com', '9003333333', '2025-06-05');
INSERT INTO Loans (book_id, member_id, loan_date, due_date, return_date) VALUES
(1, 1, date('now','-12 day'), date('now','+18 day'), NULL),
(2, 1, date('now','-40 day'), date('now','-10 day'), NULL),
(3, 2, date('now','-20 day'), date('now','+10 day'), NULL),
(4, 3, date('now','-50 day'), date('now','-20 day'), date('now','-5 day')),
(2, 3, date('now','-5 day'), date('now','+25 day'), NULL);
"""
query_currently_issued = """
SELECT
l.loan_id,
b.book_id,
b.title,
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
QUERY RESULTS
Python De
PS C:\Users\shash> c:; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:
\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '60
105' '--' 'c:\Users\shash\AAC A 16.3.py' ...
Overdue books (loan date > 30 days):
(2, '1984', 'Asha Rao', '2026-02-20', '2026-03-22', 40)
Books loaned by each member:
(1, 'Asha Rao', 2)
(3, 'Nina Paul', 2)
1 Hidden

AAC A 16.3.py

C: > Users > shash > AAC A 16.3.py > ...

```
74      l.loan_id,
75      b.book_id,
76      b.title,
77      b.author,
78      m.member_id,
79      m.full_name,
80      l.loan_date,
81      l.due_date
82  FROM Loans l
83  JOIN Books b ON b.book_id = l.book_id
84  JOIN Members m ON m.member_id = l.member_id
85  WHERE l.return_date IS NULL
86  ORDER BY l.loan_date DESC;
87  """
88
89  query_overdue_30_days = """
90  SELECT
91      l.loan_id,
92      b.title,
93      m.full_name,
94      l.loan_date,
95      l.due_date,
96      CAST(julianday('now') - julianday(l.loan_date) AS INTEGER) AS days_since_loan
97  FROM Loans l
98  JOIN Books b ON b.book id = l.book id
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

```
PS C:\Users\shash>  c:; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\pythor
\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\laur
105' '--' 'c:\Users\shash\AAC A 16.3.py' …

Books loaned by each member:
(1, 'Asha Rao', 2)
(3, 'Nina Paul', 2)
(2, 'Kiran Das', 1)
PS C:\Users\shash>
```

AAC A 16.3.py

C: > Users > shash > AAC A 16.3.py > ...

```python
def run():
    conn.executescript(schema_sql)
    conn.executescript(seed_sql)

    print("Indexing Strategy:")
    for s in indexing_strategy:
        print("-", s)

    print("\nAll books currently issued:")
    for row in conn.execute(query_currently_issued):
        print(row)

    print("\nOverdue books (loan date > 30 days):")
    for row in conn.execute(query_overdue_30_days):
        print(row)

    print("\nBooks loaned by each member:")
    for row in conn.execute(query_books_loaned_by_member):
        print(row)

    conn.close()

if __name__ == "__main__":
    run()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

```
PS C:\Users\shash>  c:; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\
\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bur
105' '--' 'c:\Users\shash\AAC A 16.3.py' …

Books loaned by each member:
(1, 'Asha Rao', 2)
(3, 'Nina Paul', 2)
(2, 'Kiran Das', 1)
PS C:\Users\shash>
```

Task 4: Optimize Queries

Instructions:

- Use AI tools to analyze query efficiency.
- Optimize inefficient queries using joins, indexes, or reducing subqueries.
- Test optimized queries for correctness.

Starter Code Example:

-- Original query

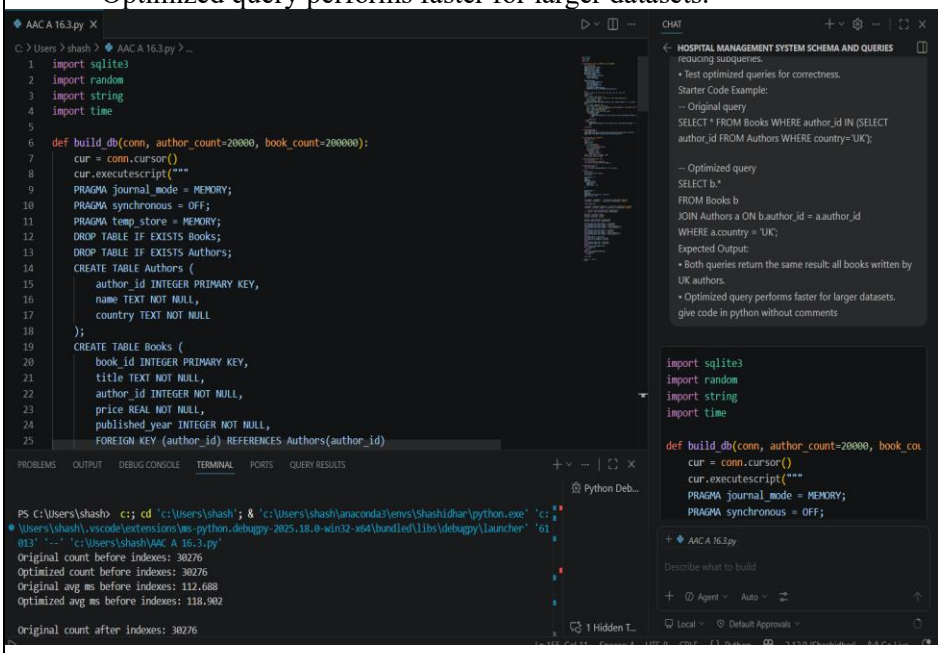
SELECT * FROM Books WHERE author_id IN (SELECT author_id FROM Authors WHERE country='UK');

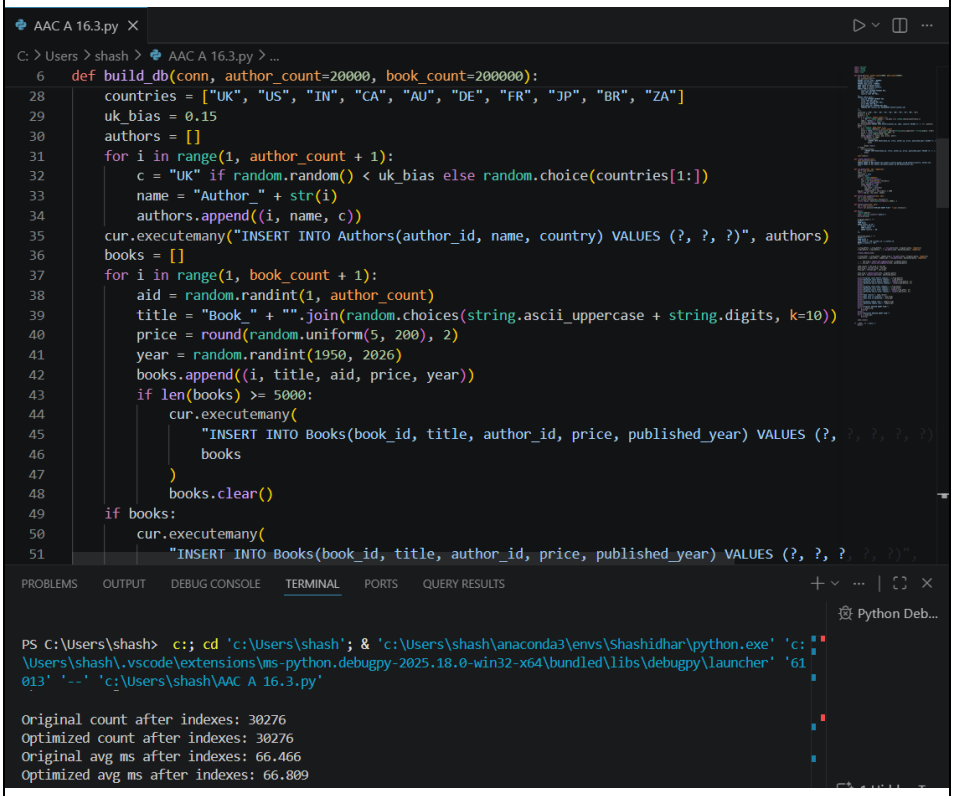
-- Optimized query

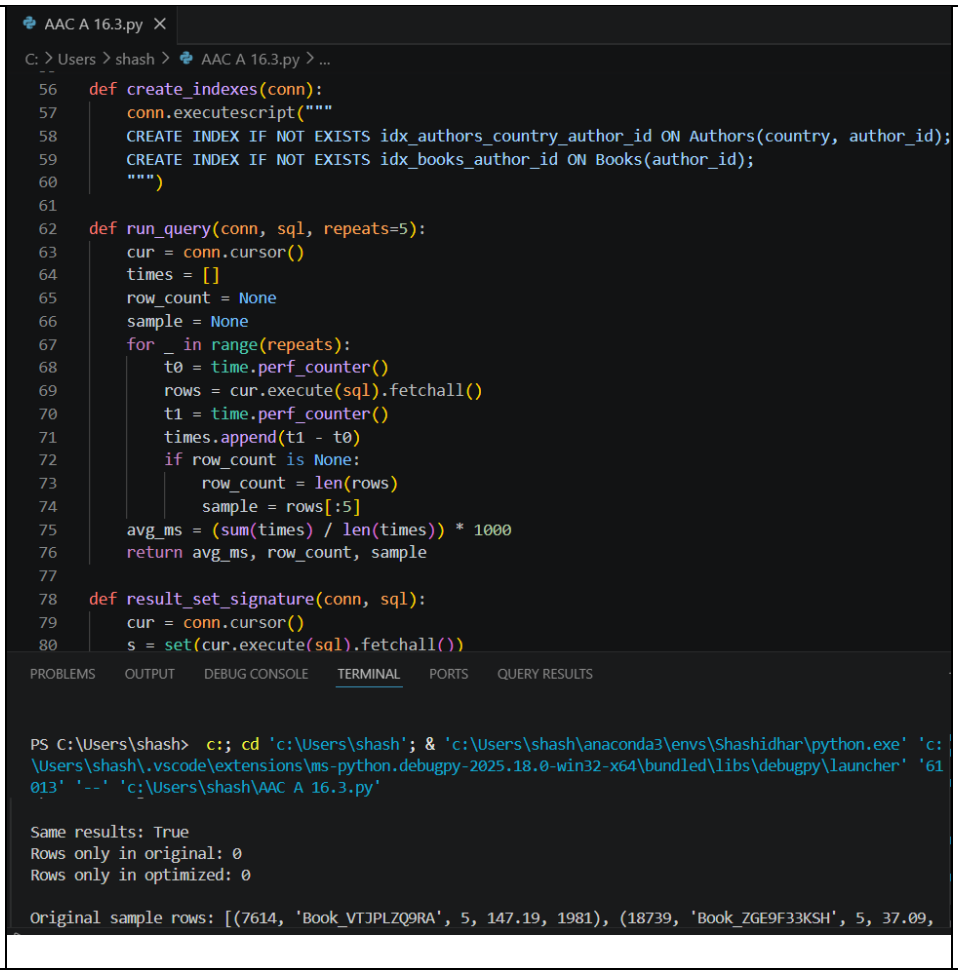
SELECT b.*
FROM Books b
JOIN Authors a ON b.author_id = a.author_id
WHERE a.country = 'UK';

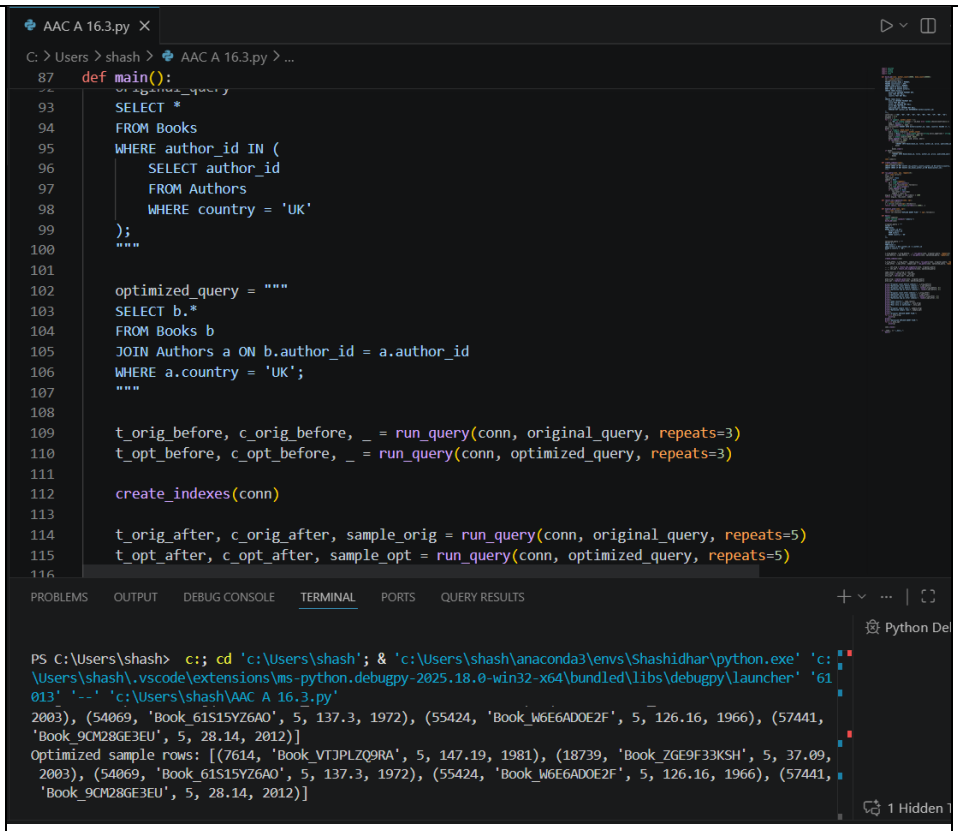
Expected Output:

- Both queries return the same result: all books written by UK authors.
- Optimized query performs faster for larger datasets.






AAC A 16.3.py
C: > Users > shash > AAC A 16.3.py > ...
56 def create_indexes(conn):
57 conn.executescript("""
58 CREATE INDEX IF NOT EXISTS idx_authors_country_author_id ON Authors(country, author_id);
59 CREATE INDEX IF NOT EXISTS idx_books_author_id ON Books(author_id);
60 """)
61
62 def run_query(conn, sql, repeats=5):
63 cur = conn.cursor()
64 times = []
65 row_count = None
66 sample = None
67 for _ in range(repeats):
68 t0 = time.perf_counter()
69 rows = cur.execute(sql).fetchall()
70 t1 = time.perf_counter()
71 times.append(t1 - t0)
72 if row_count is None:
73 row_count = len(rows)
74 sample = rows[:5]
75 avg_ms = (sum(times) / len(times)) * 1000
76 return avg_ms, row_count, sample
77
78 def result_set_signature(conn, sql):
79 cur = conn.cursor()
80 s = set(cur.execute(sql).fetchall())
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS
PS C:\Users\shash> c:; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '61013' '--' 'c:\Users\shash\AAC A 16.3.py'
Same results: True
Rows only in original: 0
Rows only in optimized: 0
Original sample rows: [(7614, 'Book_VTJPLZQ9RA', 5, 147.19, 1981), (18739, 'Book_ZGE9F33KSH', 5, 37.09,


AAC A 16.3.py
C: > Users > shash > AAC A 16.3.py > ...
def main():
SELECT *
FROM Books
WHERE author_id IN (
SELECT author_id
FROM Authors
WHERE country = 'UK'
);
"""
optimized_query = """
SELECT b.*
FROM Books b
JOIN Authors a ON b.author_id = a.author_id
WHERE a.country = 'UK';
"""
t_orig_before, c_orig_before, _ = run_query(conn, original_query, repeats=3)
t_opt_before, c_opt_before, _ = run_query(conn, optimized_query, repeats=3)
create_indexes(conn)
t_orig_after, c_orig_after, sample_orig = run_query(conn, original_query, repeats=5)
t_opt_after, c_opt_after, sample_opt = run_query(conn, optimized_query, repeats=5)
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
QUERY RESULTS
Python Del
PS C:\Users\shash> c:; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '61013' '--' 'c:\Users\shash\AAC A 16.3.py'
2003), (54069, 'Book_61S15YZ6AO', 5, 137.3, 1972), (55424, 'Book_W6E6ADOE2F', 5, 126.16, 1966), (57441, 'Book_9CM28GE3EU', 5, 28.14, 2012)]
Optimized sample rows: [(7614, 'Book_VTJPLZQ9RA', 5, 147.19, 1981), (18739, 'Book_ZGE9F33KSH', 5, 37.09, 2003), (54069, 'Book_61S15YZ6AO', 5, 137.3, 1972), (55424, 'Book_W6E6ADOE2F', 5, 126.16, 1966), (57441, 'Book_9CM28GE3EU', 5, 28.14, 2012)]
1 Hidden

AAC A 16.3.py

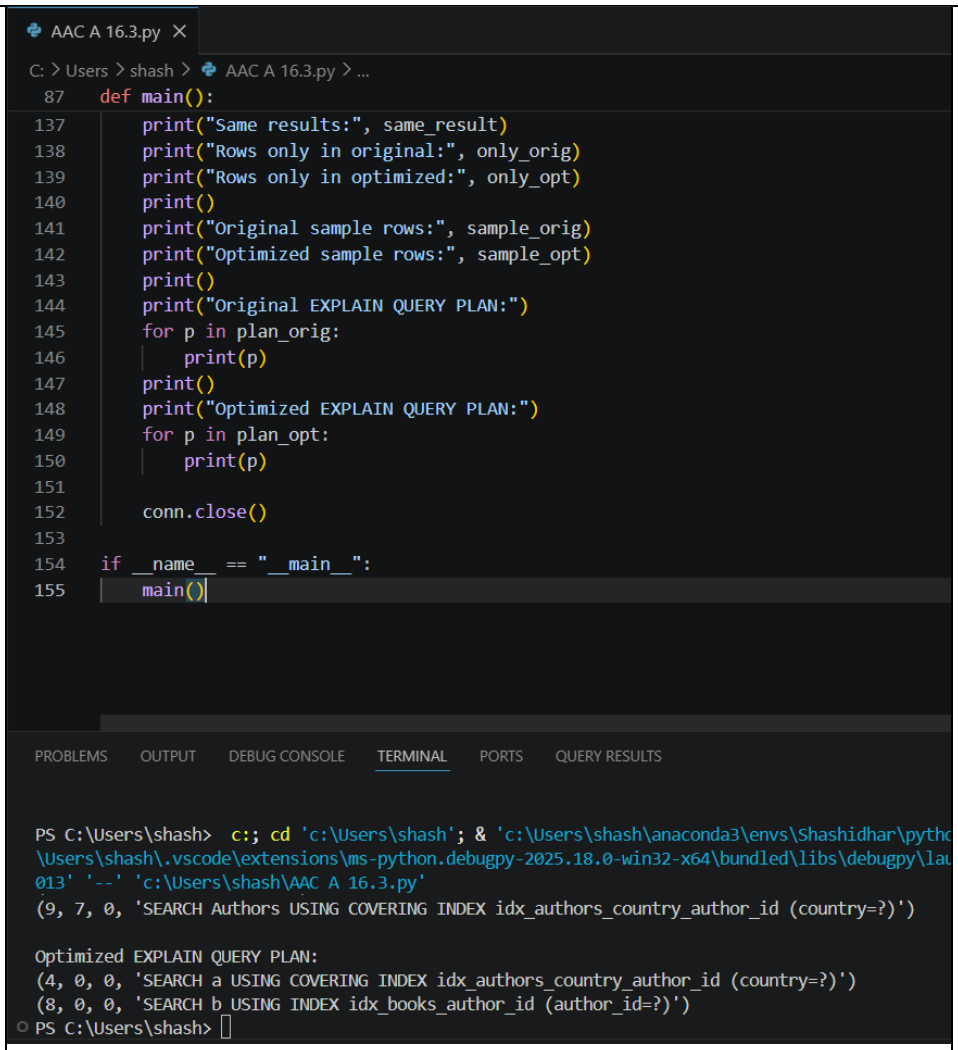
C: > Users > shash > AAC A 16.3.py > ...

```
 87  def main():
120      same_result = set_orig == set_opt
121      only_orig = len(set_orig - set_opt)
122      only_opt = len(set_opt - set_orig)
123
124      plan_orig = explain_plan(conn, original_query)
125      plan_opt = explain_plan(conn, optimized_query)
126
127      print("Original count before indexes:", c_orig_before)
128      print("Optimized count before indexes:", c_opt_before)
129      print("Original avg ms before indexes:", round(t_orig_before, 3))
130      print("Optimized avg ms before indexes:", round(t_opt_before, 3))
131      print()
132      print("Original count after indexes:", c_orig_after)
133      print("Optimized count after indexes:", c_opt_after)
134      print("Original avg ms after indexes:", round(t_orig_after, 3))
135      print("Optimized avg ms after indexes:", round(t_opt_after, 3))
136      print()
137      print("Same results:", same_result)
138      print("Rows only in original:", only_orig)
139      print("Rows only in optimized:", only_opt)
140      print()
141      print("Original sample rows:", sample_orig)
142      print("Optimized sample rows:", sample_opt)
143      print()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

```
PS C:\Users\shash>  c:; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.ex
\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launche
013' '--' 'c:\Users\shash\AAC A 16.3.py'

Original EXPLAIN QUERY PLAN:
(3, 0, 0, 'SEARCH Books USING INDEX idx_books_author_id (author_id=?)')
(7, 0, 0, 'LIST SUBQUERY 1')
(9, 7, 0, 'SEARCH Authors USING COVERING INDEX idx_authors_country_author_id (country=?)')
```

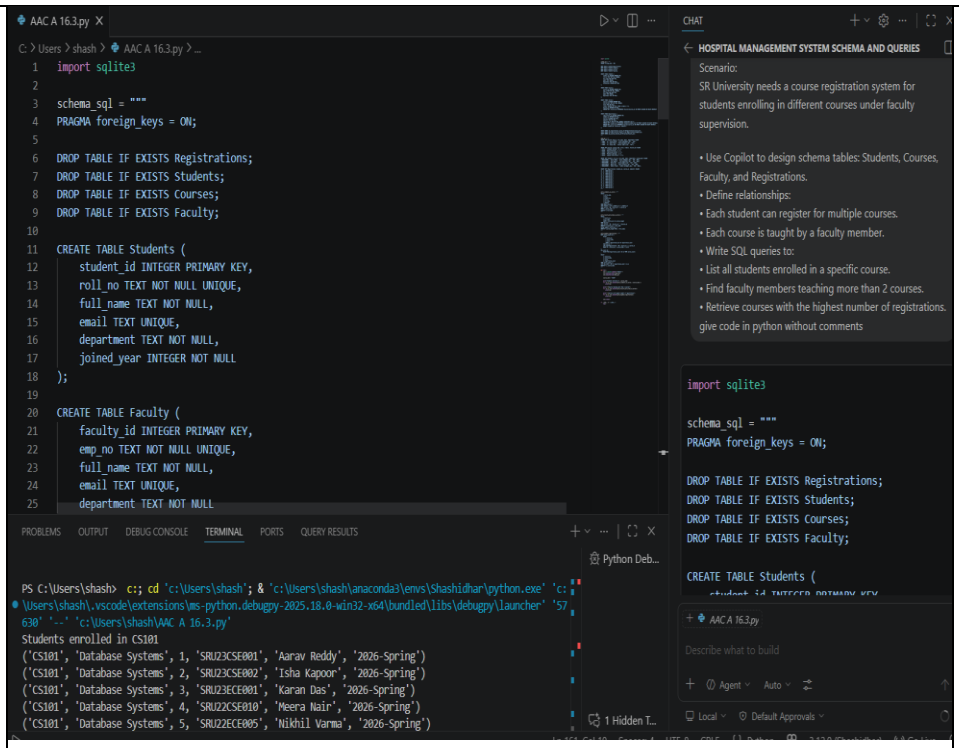


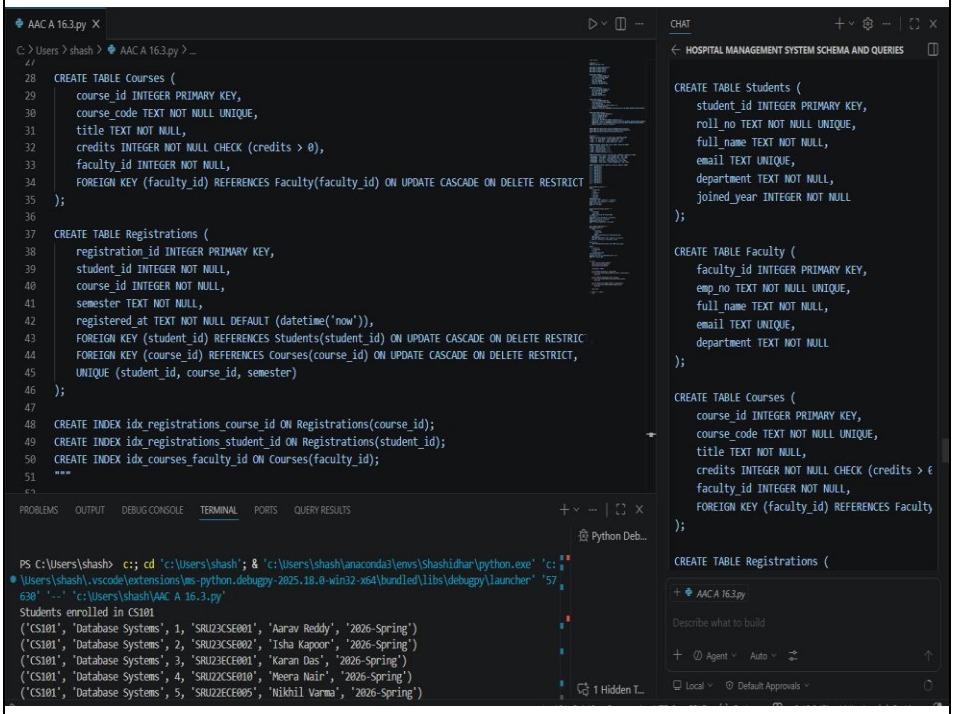
Task 5: University Course Registration

Scenario:

SR University needs a **course registration system** for students enrolling in different courses under faculty supervision.

- Use Copilot to design schema tables: Students, Courses, Faculty, and Registrations.
- Define relationships:
  - Each student can register for multiple courses.
  - Each course is taught by a faculty member.
- Write SQL queries to:
  - List all students enrolled in a specific course.
  - Find faculty members teaching more than 2 courses.
  - Retrieve courses with the highest number of registrations.


AAC A 16.3.py
C: > Users > shash > AAC A 16.3.py > ...
import sqlite3
schema_sql = """
PRAGMA foreign_keys = ON;
DROP TABLE IF EXISTS Registrations;
DROP TABLE IF EXISTS Students;
DROP TABLE IF EXISTS Courses;
DROP TABLE IF EXISTS Faculty;
CREATE TABLE Students (
student_id INTEGER PRIMARY KEY,
roll_no TEXT NOT NULL UNIQUE,
full_name TEXT NOT NULL,
email TEXT UNIQUE,
department TEXT NOT NULL,
joined_year INTEGER NOT NULL
);
CREATE TABLE Faculty (
faculty_id INTEGER PRIMARY KEY,
emp_no TEXT NOT NULL UNIQUE,
full_name TEXT NOT NULL,
email TEXT UNIQUE,
department TEXT NOT NULL
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS
Python Deb...
PS C:\Users\shash> c:; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57630' '--' 'c:\Users\shash\AAC A 16.3.py'
Students enrolled in CS101
('CS101', 'Database Systems', 1, 'SRU23CSE001', 'Aarav Reddy', '2026-Spring')
('CS101', 'Database Systems', 2, 'SRU23CSE002', 'Isha Kapoor', '2026-Spring')
('CS101', 'Database Systems', 3, 'SRU23ECE001', 'Karan Das', '2026-Spring')
('CS101', 'Database Systems', 4, 'SRU22CSE010', 'Meera Nair', '2026-Spring')
('CS101', 'Database Systems', 5, 'SRU22ECE005', 'Nikhil Varma', '2026-Spring')
1 Hidden T...
CHAT
HOSPITAL MANAGEMENT SYSTEM SCHEMA AND QUERIES
Scenario:
SR University needs a course registration system for students enrolling in different courses under faculty supervision.
• Use Copilot to design schema tables: Students, Courses, Faculty, and Registrations.
• Define relationships:
• Each student can register for multiple courses.
• Each course is taught by a faculty member.
• Write SQL queries to:
• List all students enrolled in a specific course.
• Find faculty members teaching more than 2 courses.
• Retrieve courses with the highest number of registrations.
give code in python without comments
import sqlite3
schema_sql = """
PRAGMA foreign_keys = ON;
DROP TABLE IF EXISTS Registrations;
DROP TABLE IF EXISTS Students;
DROP TABLE IF EXISTS Courses;
DROP TABLE IF EXISTS Faculty;
CREATE TABLE Students (
AAC A 16.3.py
Describe what to build
Agent
Auto
Local
Default Approvals


AAC A 16.3.py
C: > Users > shash > AAC A 16.3.py > ...
CREATE TABLE Courses (
course_id INTEGER PRIMARY KEY,
course_code TEXT NOT NULL UNIQUE,
title TEXT NOT NULL,
credits INTEGER NOT NULL CHECK (credits > 0),
faculty_id INTEGER NOT NULL,
FOREIGN KEY (faculty_id) REFERENCES Faculty(faculty_id) ON UPDATE CASCADE ON DELETE RESTRICT
);
CREATE TABLE Registrations (
registration_id INTEGER PRIMARY KEY,
student_id INTEGER NOT NULL,
course_id INTEGER NOT NULL,
semester TEXT NOT NULL,
registered_at TEXT NOT NULL DEFAULT (datetime('now')),
FOREIGN KEY (student_id) REFERENCES Students(student_id) ON UPDATE CASCADE ON DELETE RESTRIC
FOREIGN KEY (course_id) REFERENCES Courses(course_id) ON UPDATE CASCADE ON DELETE RESTRICT,
UNIQUE (student_id, course_id, semester)
);
CREATE INDEX idx_registrations_course_id ON Registrations(course_id);
CREATE INDEX idx_registrations_student_id ON Registrations(student_id);
CREATE INDEX idx_courses_faculty_id ON Courses(faculty_id);
"""
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS
Python Deb...
PS C:\Users\shash> c:; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57630' '--' 'c:\Users\shash\AAC A 16.3.py'
Students enrolled in CS101
('CS101', 'Database Systems', 1, 'SRU23CSE001', 'Aarav Reddy', '2026-Spring')
('CS101', 'Database Systems', 2, 'SRU23CSE002', 'Isha Kapoor', '2026-Spring')
('CS101', 'Database Systems', 3, 'SRU23ECE001', 'Karan Das', '2026-Spring')
('CS101', 'Database Systems', 4, 'SRU22CSE010', 'Meera Nair', '2026-Spring')
('CS101', 'Database Systems', 5, 'SRU22ECE005', 'Nikhil Varma', '2026-Spring')
1 Hidden T...
CHAT
HOSPITAL MANAGEMENT SYSTEM SCHEMA AND QUERIES
CREATE TABLE Students (
student_id INTEGER PRIMARY KEY,
roll_no TEXT NOT NULL UNIQUE,
full_name TEXT NOT NULL,
email TEXT UNIQUE,
department TEXT NOT NULL,
joined_year INTEGER NOT NULL
);
CREATE TABLE Faculty (
faculty_id INTEGER PRIMARY KEY,
emp_no TEXT NOT NULL UNIQUE,
full_name TEXT NOT NULL,
email TEXT UNIQUE,
department TEXT NOT NULL
);
CREATE TABLE Courses (
course_id INTEGER PRIMARY KEY,
course_code TEXT NOT NULL UNIQUE,
title TEXT NOT NULL,
credits INTEGER NOT NULL CHECK (credits > 0
faculty_id INTEGER NOT NULL,
FOREIGN KEY (faculty_id) REFERENCES Faculty
);
CREATE TABLE Registrations (
AAC A 16.3.py
Describe what to build
Agent
Auto
Local
Default Approvals

AAC A 16.3.py

C: > Users > shash > AAC A 16.3.py > ...

```
48  CREATE INDEX idx_registrations_course_id ON Registrations(course_id);
49  CREATE INDEX idx_registrations_student_id ON Registrations(student_id);
50  CREATE INDEX idx_courses_faculty_id ON Courses(faculty_id);
51  """
52
53  seed_sql = """
54  INSERT INTO Faculty (emp_no, full_name, email, department) VALUES
55  ('F1001', 'Dr. Priya Sharma', 'priya.sharma@sru.edu', 'CSE'),
56  ('F1002', 'Dr. Rahul Menon', 'rahul.menon@sru.edu', 'CSE'),
57  ('F1003', 'Dr. Ananya Rao', 'ananya.rao@sru.edu', 'ECE');
58
59  INSERT INTO Courses (course_code, title, credits, faculty_id) VALUES
60  ('CS101', 'Database Systems', 4, 1),
61  ('CS102', 'Data Structures', 4, 1),
62  ('CS201', 'Operating Systems', 4, 1),
63  ('CS202', 'Machine Learning', 3, 2),
64  ('EC101', 'Digital Electronics', 4, 3);
65
66  INSERT INTO Students (roll_no, full_name, email, department, joined_year) VALUES
67  ('SRU23CSE001', 'Aarav Reddy', 'aarav.reddy@sru.edu', 'CSE', 2023),
68  ('SRU23CSE002', 'Isha Kapoor', 'isha.kapoor@sru.edu', 'CSE', 2023),
69  ('SRU23ECE001', 'Karan Das', 'karan.das@sru.edu', 'ECE', 2023),
70  ('SRU22CSE010', 'Meera Nair', 'meera.nair@sru.edu', 'CSE', 2022),
71  ('SRU22ECE005', 'Nikhil Varma', 'nikhil.varma@sru.edu', 'ECE', 2022);
72
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

```
PS C:\Users\shash>  c:; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe'
\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher'
630' '--' 'c:\Users\shash\AAC A 16.3.py'

Faculty teaching more than 2 courses
(1, 'Dr. Priya Sharma', 3)

Courses with highest number of registrations
(1, 'CS101', 'Database Systems', 5)
```

AAC A 16.3.py

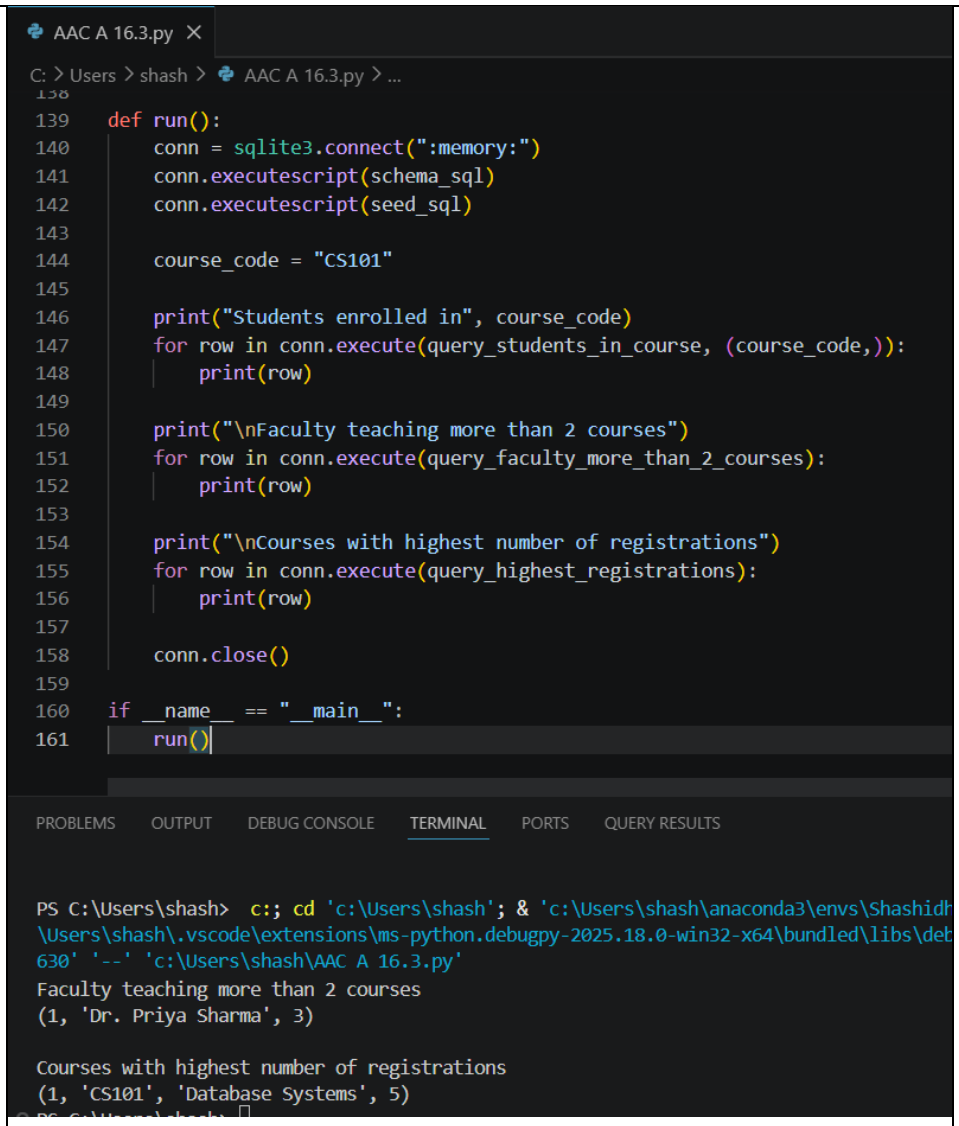
C: > Users > shash > AAC A 16.3.py > ...

```
73    INSERT INTO Registrations (student_id, course_id, semester) VALUES
74    (1, 1, '2026-Spring'),
75    (1, 2, '2026-Spring'),
76    (1, 4, '2026-Spring'),
77    (2, 1, '2026-Spring'),
78    (2, 3, '2026-Spring'),
79    (3, 5, '2026-Spring'),
80    (3, 1, '2026-Spring'),
81    (4, 1, '2026-Spring'),
82    (4, 2, '2026-Spring'),
83    (4, 3, '2026-Spring'),
84    (5, 5, '2026-Spring'),
85    (5, 1, '2026-Spring');
86    """
87
88    query_students_in_course = """
89    SELECT
90        c.course_code,
91        c.title,
92        s.student_id,
93        s.roll_no,
94        s.full_name,
95        r.semester
96    FROM Registrations r
97    JOIN Students s ON s.student_id = r.student_id
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS

```
PS C:\Users\shash>  c:; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\Sha
\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\lib
630' '--' 'c:\Users\shash\AAC A 16.3.py'
Faculty teaching more than 2 courses
(1, 'Dr. Priya Sharma', 3)

Courses with highest number of registrations
(1, 'CS101', 'Database Systems', 5)
```



Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.